Who is on the other side of your trade?

How exchange rules distribute trading costs and risks, and what this means for protecting everyday traders

Rent or toxicity: strategic exploitability of learning market makers with public wallet information

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6.S974 Games, Learning, and Security, Workshop 1
Example 4: an algorithmic market with adaptive agents

2026: the year retail leverage went wrong at national scale

−44%

KOSPI, June 19 to July 29

1.2 million

retail accounts margin-called:
3.4% of all adults

$39 billion

retail losses on single-stock leveraged ETFs

“Retail” means everyday people

- 92% of the holders were individuals, not firms
- 62% of the accounts: people aged 20 to 39
- taking leveraged exposure through products whose risks can be hard to assess

The public-benefit goal

Help people avoid preventable trading losses.

Today: identify mechanisms that distribute trading costs and risks.

Next: test whether changing the rules improves ordinary traders’ outcomes.

Sources for background figures: Bloomberg, Reuters, Goldman Sachs, Citi (via Seoulz, “KOSPI Crash 2026”); Maeil Shinmun, Oct 22, 2026.

Leverage magnifies losses; liquidation is a future extension

You put in $1,000. At $20\times$ leverage you control a $20,000 position.



collateral: $1,000 **exposure beyond collateral**: $19,000

price $+5\%$

position gain: $20,000 $\times$ 5%



you doubled your money: $+$1,000

price -5%

position loss: $20,000 $\times$ 5%



loss equals collateral: $-$1,000

illustrative PnL before costs; maintenance margin can trigger liquidation before this point

Current model: quotes, fills, information and queue position. It does not track leveraged accounts.

Later extension: collateral, margin and forced sales, to test protections against liquidation losses.

From trading mechanisms to protections we can test

What this study establishes

- learned maker profits can be vulnerable to a strategic entrant
- wallet information can help select fills in the model
- identity and priority rules change who bears toxic flow

The missing test

Does changing a rule improve outcomes for traders with less information and fewer resources?

Public wallets make the question measurable

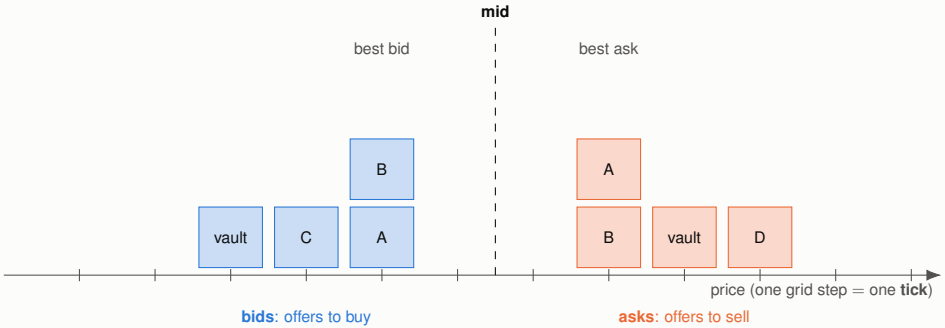
fill buy 0.5 BTC @ 67,412.0
taker `0x7f3a...c91e` maker `0x21b0...4d7a`
a wallet's history is observable

Next: measure customer outcomes

- execution costs and gains from trade
- access to liquidity during stress
- severe losses for passive traders
- whether improvements survive adaptation

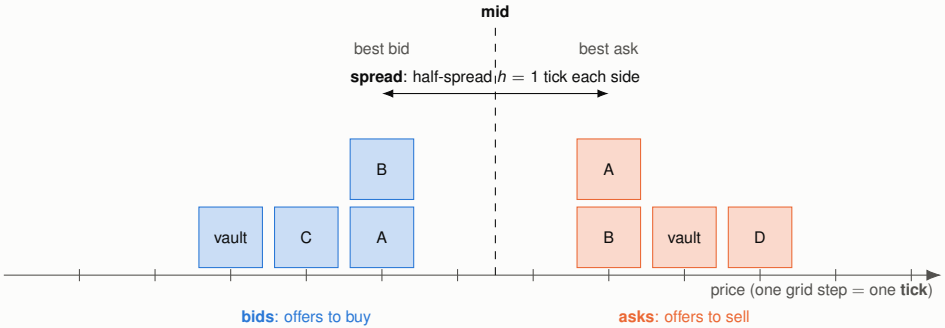
Consumer protection is the next test. Current evidence identifies mechanisms and tradeoffs.

An exchange is a list of standing offers: the order book



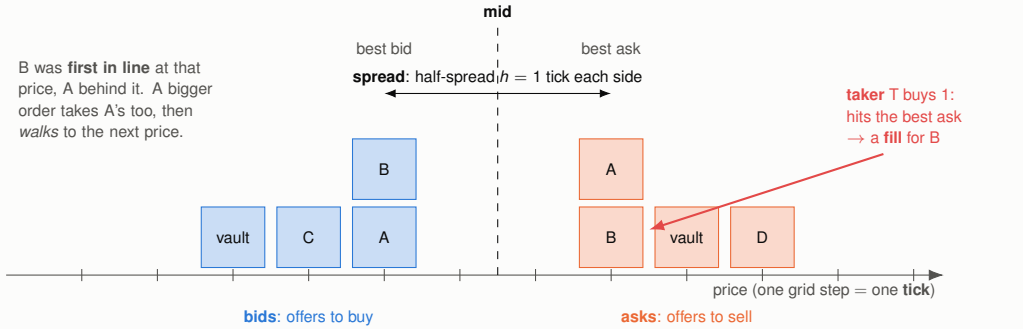
Makers post the offers (the boxes). **Takers** accept them.
Each box carries a public **wallet** address.

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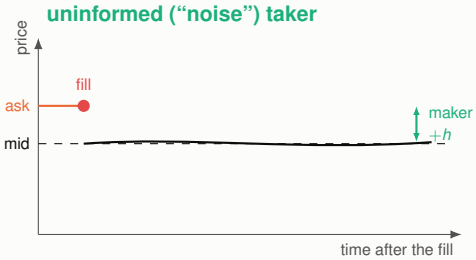
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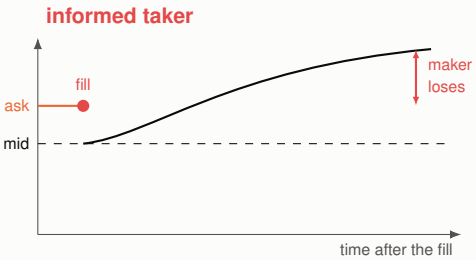
Makers post the offers (the boxes). **Takers** accept them. Each box carries a public **wallet** address.

Public, at once: buy, price, wallet T, wallet B.
Public, only later: whether B gained or lost on it.

Did the maker win? Look at the price afterwards: the markout



price does not move; the maker keeps the half-spread

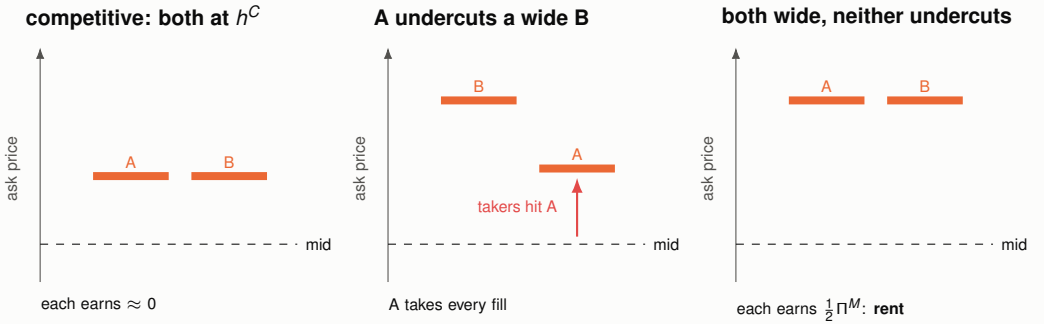


the taker knew; the price runs past the fill price

markout = mid a few seconds later − fill price, signed for the maker

toxic flow: a stream of fills with negative markouts (*adverse selection*)

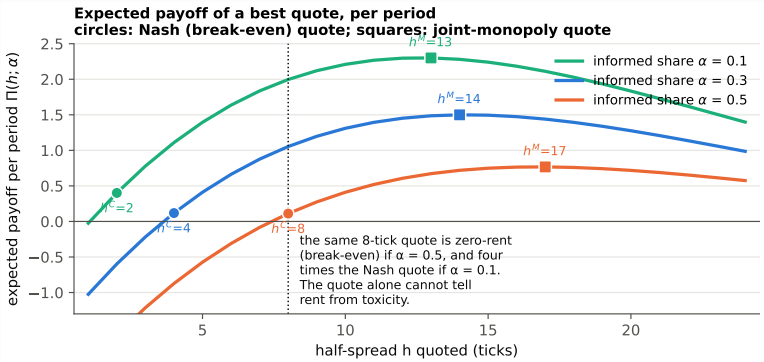
Rent: what makers earn by not undercutting each other



h^C: the **competitive** quote, break-even. Wider pays, but invites undercutting.

Learning makers end up in the right panel *without talking*: **tacit collusion**. The extra is the **rent**.

The puzzle: on the tape, rent and toxicity look identical



A wide quote is a **rent** . . . or the correct charge for **toxic flow**.

Which curve you are on is decided by α, the share of informed takers. Only outcomes, or *who* trades, reveal it.

Three players, and what each one gets to know

incumbent makers
learned a cartel; decide *how wide*, and whether to stay in line

an entrant
decides which quotes to undercut, which fills to be there for

the venue
decides the rules: public identities? tick size? pay for position?

the entrant's information set is the *treatment*



What we want to learn

RQ1 Value of identity

What is a public wallet worth to an entrant, beyond the tape, and through which mechanism?

RQ2 Exploitability

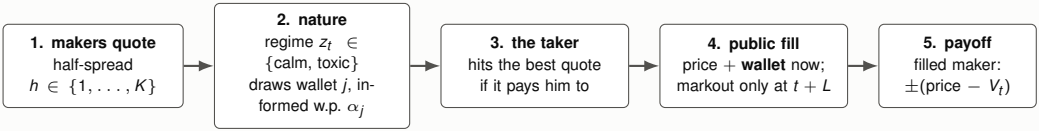
How much of a learned cartel's rent can an adversary take or destroy?

RQ3 One-tick books

When the grid pins the spread, where does rent live, and which rules decide who holds the toxic flow?

Today: identify mechanisms. Next: test rules by customer outcomes.

One period of the game



Incumbents: tabular Q-learners, state = (own last h, rival's last h); trained, then **frozen**.

Entrant: information set $\mathcal{F}^{\text{anon}} \subset \mathcal{F}^{\text{id}} \subset \mathcal{F}^{\text{oracle}}$; a Bayesian filter estimates the next taker's α.

Payoff curve, closed form:
$\Pi(h; \alpha) =$
$(1 - \alpha)\, 2x\bar{\Phi}(x/\sigma_L) - \alpha\mu e^{-x/\mu}$,
$x = h\theta$.

Benchmarks are closed-form, so “rent” is measured, not estimated

$h^C = \min\{h : \Pi(h) \geq 0\}$	**Nash** (competitive)
$h^M = \arg\max_h \Pi(h)$	**joint monopoly**
$C = \dfrac{\Pi_{\text{cartel}} - \Pi^C}{\Pi^M - \Pi^C} \in [0,1]$	**rent index**
$L_{\text{BR}} = \Pi_{\text{cartel}} - \Pi_{\text{under attack}}$	**loss under attack**
$\kappa_A = L_{\text{BR}}/(\Pi_{\text{cartel}} - \Pi^C)$	**rent destroyed**

At $\bar{\alpha} = 0.3$: $h^C = 4$, $h^M = 14$, rent $\Pi^M - \Pi^C = 1.38$.

Why a cartel is attackable: it is a *reaction function* that does not chase every undercut. That restraint is the rent, and the attack surface.

One-tick (queue) mode: spread pinned; actions REST / OUT / REST-AND-PAY, first in first out. A fill at h^C: $+1.10$ calm, -2.26 toxic.

The prototype, in one command

`python experiments/demo.py`

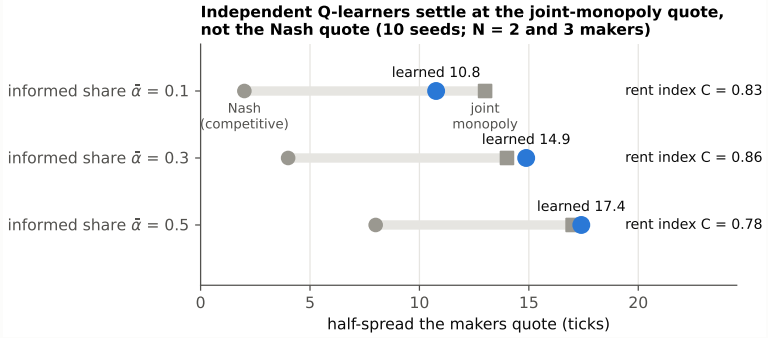
1. benchmarks h^C, h^M, $\Pi(h)$
2. the shipped cartel alone: $C = 0.86$
3. three adversaries enter
4. the best response, period by period

`github.com/pradeepu191/wargame`
59 tests; every result logged with its command.

```
Entry (200 episodes each, same random numbers):
  adversary                          cartel loses  entrant earns  kappa_A
  undercut (one tick inside)               0.41           0.33     0.48
  markout (undercut if rent inferred)      0.47           0.41     0.55
  exact best response                      0.94           1.09     1.10

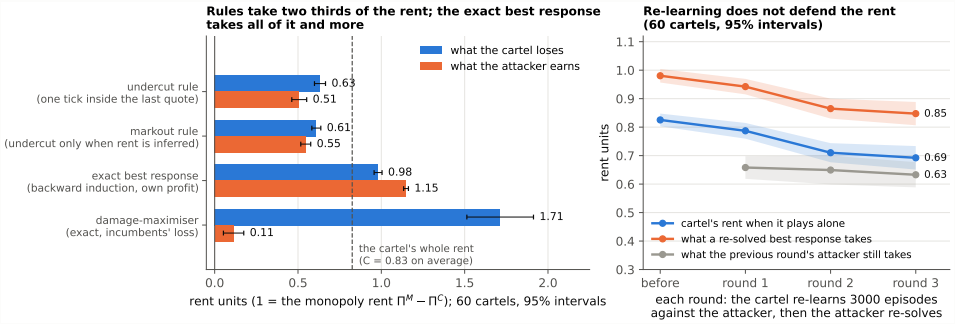
Trace: exact best response vs the cartel (quotes in ticks)
   t  inc0  inc1  entrant   taker          filled   payoff
   0    15    24       12   noise     buy  entrant   +1.21
   1    23    16       13   noise     buy  entrant   +7.86
   2    20    14       13   no trade
   5    21    19       14   informed buy   entrant   -5.77
   7    18    23       14   noise     buy  entrant  +11.79
```

Evidence 1: learners coordinate above the Nash point



$C = 0.71$ to 0.87 in every cell. In 55 to 83% of states undercutting would pay *this period*; the learner forgoes it 96 to 100% of the time.

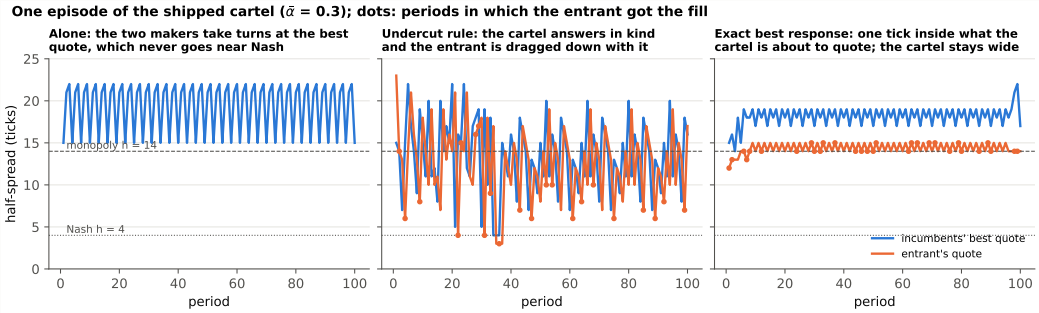
Evidence 2: the rent is fully exploitable



Hand-written rules take two thirds. The **exact best response** takes it all, in 60 of 60 cartels: $\kappa_A = 1.1$ to 1.3.

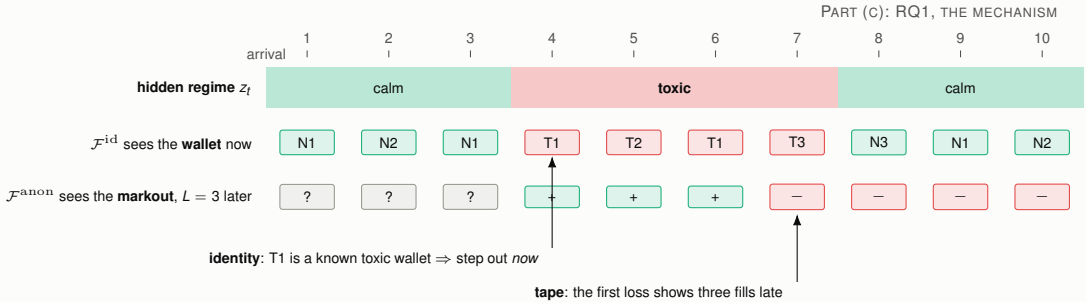
The damage-maximiser destroys 2 to 3$\times$ the rent at no cost.

Steering, period by period



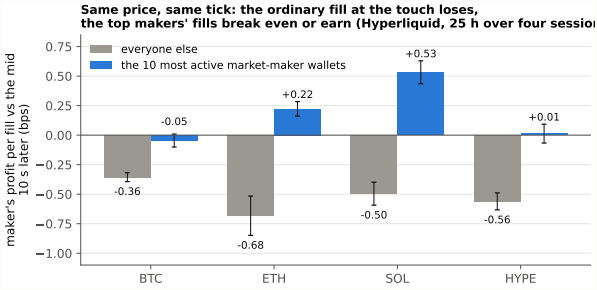
Left: the cartel alone rotates at wide quotes. **Middle**: a naive undercut starts a price war. **Right**: the best response holds the cartel wide and takes the fills.

Why identity has value: a spell of toxic flow, seen two ways



Identity is worth what you lose in the L fills before the tape catches up: measured, zero at $\bar{\alpha} \leq 0.3$, growing with L at $\bar{\alpha} = 0.5$ (backup).

Evidence 3, Hyperliquid: the price cannot move, and someone still earns

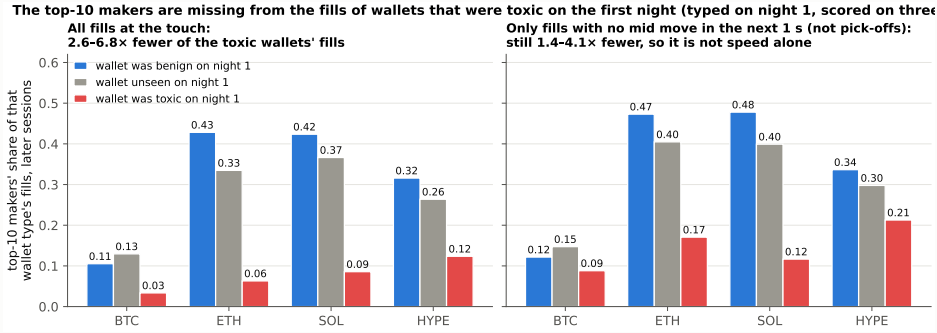


Four sessions, 25 h, 490k scored fills.

- **BTC/ETH/SOL spreads**: essentially one tick
- **Average maker markout**: negative at 10 s on every coin
- **The top makers earn** $+0.3$ to $+1.0$ bps more, *at the same price*; rank persists (corr. 0.5 to 0.8)

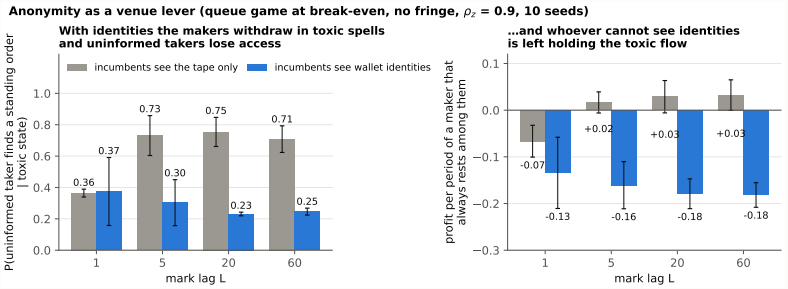
Fill selection matters even at the same quoted price.

Evidence 4: top makers receive fewer toxic-wallet fills



Wallets typed on night 1, scored on three later sessions. Consistent with selection; the makers' decision signals remain unobserved.

Evidence 5: venue rules redistribute access and losses



Anonymity: more uninformed takers get filled in stress in the queue model.

Priority fee: venue revenue rises; the always-resting fringe's losses deepen.

Next test: customer costs and severe losses, alongside access and maker viability.

Workshop 2: test protections for less-informed traders

1 Customer outcomes

Separate small **takers** and **passive traders**: execution costs, gains from trade, stress access and worst-5% losses.

2 Compare rules

Public, anonymous or delayed wallet disclosure. FIFO versus paid priority. Measure **who gains and who loses**.

3 Stress-test benefits

Let makers adapt. Use the existing attacker as a stress test. Check liquidity and whether loss-making participants leave.

Your view: which rule improves less-informed traders' outcomes, and does the improvement survive stress and adaptation?

Longer term: connect execution risk to liquidation losses

Add to the model

- persistent prices and finite book depth
- account collateral and maintenance margin
- forced sales and their price impact

Compare: leverage limits, margin buffers and liquidation execution rules.

Measure protection directly

- fraction of small accounts liquidated
- collateral remaining after liquidation
- severe losses and ability to exit in stress

Evidence: longer recording windows and stress episodes to calibrate the extension.

> **Reducing liquidation losses is a future hypothesis. Today's model cannot establish it.**

If you remember three things

1 Vulnerabilities

Learned maker profits are exploitable in our model. This gives us a way to stress-test market rules.

2 Distribution

Information and priority change who gets fills and who bears losses. The data motivates testing these mechanisms.

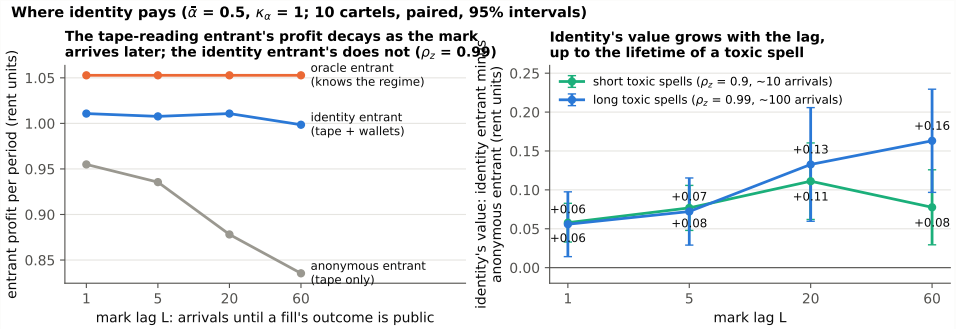
3 Protection next

Test rules by customer costs, access and severe losses. Measure whether less-informed traders benefit.

Goal: fewer preventable losses for everyday traders.

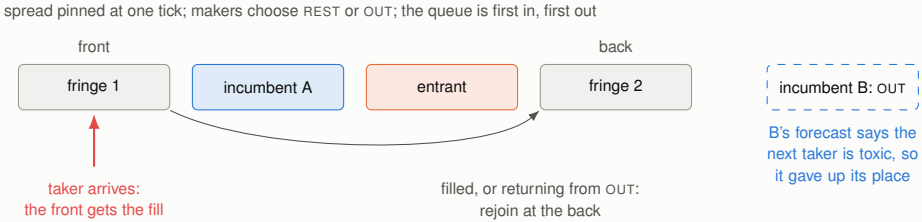
`github.com/pradeepu191/wargame`

Backup: what identity is worth, measured



The proposal's mechanism (the same wallet trades again) is worth nothing: repeats are 2 to 5%. The lag mechanism survives: zero where $\bar{\alpha} \leq 0.3$, growing with L up to the spell's lifetime.

Backup: the queue game, the one-tick book



A fill at h^C: $+1.10$ in a calm spell, -2.26 in a toxic one.

Who pays in the model: the always-resting *fringe*. Its retail interpretation needs testing.

Backup: Hyperliquid numbers

	BTC	ETH	SOL	HYPE
one tick (bps)	0.12	0.39	0.86	0.11
taker's cost vs the mid: fills at the touch / walked (bps)	0.12 / 2.35	0.42 / 3.20	0.92 / 5.20	0.17 / 3.57
10-s maker markout, at the touch / walked (bps)	-0.41 / -0.80	-0.56 / -2.26	-0.27 / $+0.98$	-0.45 / -0.01
next fill's markout after a toxic vs a benign wallet (bps, 327 h)	-0.64	-1.01	-0.59	-0.59
10-s markout at the touch: top-10 makers / everyone else (bps)	-0.05 / -0.36	$+0.22$ / -0.68	$+0.53$ / -0.50	$+0.01$ / -0.56
per-maker rank correlation, first session vs later (n makers)	0.80 (113)	0.53 (80)	0.70 (53)	0.59 (70)
top-10 share of touch fills: benign / unseen / toxic wallets	.11 / .13 / .03	.43 / .34 / .06	.42 / .37 / .09	.32 / .26 / .12
. . . on fills with no mid move in 1 s: benign / toxic	.12 / .09	.47 / .17	.48 / .12	.34 / .21

Four book sessions over two weeks (25 h). Standard errors cluster by five-minute block; every top-10-vs-rest gap exceeds four SEs. Wallet types from the first night, scored on the three later sessions.

Backup: the double oracle and robustness, in numbers

- **Double oracle** (cartel re-learns 3000 episodes against the fixed best response, entrant re-solves; 3 rounds): the now-stale attacker's profit falls 1.2 $\rightarrow$ 0.5 (the undercut-rule level); the re-solved one takes $\geq$ 90% of the remaining rent in every cartel every round; the cartel's own rent erodes 0.83 $\rightarrow$ 0.79 $\rightarrow$ 0.71 $\rightarrow$ 0.69.
- **Damage-maximiser** ($\lambda = 0$): steers the cartel to 4 to 7 ticks, inside break-even ($h^C = 8$ to 13), by quoting above it; takes no fills, earns ≈ 0; $L_{\mathrm{BR}} = 1.3$ ($\bar{\alpha} = 0.3$), 2.7 to 2.8 ($\bar{\alpha} = 0.5$).
- **Train at $\bar{\alpha}$, test at $\bar{\alpha}'$**: cartels keep $C = 0.80$ to 0.86, $\kappa_A = 0.65$ to 0.75 (rules) on calmer markets; on a more toxic one the rent collapses ($C = 0.23$) and is more than destroyed.
- **Held-out adversaries** preserve every defence ordering (rank correlation 0.75 to 0.96).
- **Caveat we own:** the exact best response needs the cartel's reaction function (oracle knowledge of the *policy*). It is the true upper bound; the public-information attacker that identifies it is next.

Backup: the queue game and the priority fee

- Learned incumbents need a **fringe** of always-resting makers (else a toxic spell empties the book for good); tabular Q cannot learn to queue (time priority pays only after consecutive periods in line); **no-regret (EXP3)** incumbents over complete policies learn to select in a few hundred episodes.
- Against selecting incumbents identity is worth *more*: $+0.05$ to $+0.06$ per period at every lag, 5/5 seeds; the fringe's payoff falls to -0.02 to -0.10: identity-blind resting liquidity is the counterparty of the selection rent.
- **Priority fee**: every selector pays in calm periods (identity-aware incumbents 93 to 94% of calm, 8 to 9% of toxic periods); a non-paying entrant among payers gets 0% of fills; the fringe's loss triples at any fee; revenue peaks at $f = 0.5$.
- **Makers' levers** (20 cartels, 7 adversaries incl. a Q-learning one): requoting every 2 to 5 periods costs everything; randomising costs profit and buys nothing; the identity floor: worst-case profit 0.08 $\rightarrow$ 0.19 at $\bar{\alpha} = 0.5$.